# Citizen Grievance & Sentiment Analysis System

SUBMITTED BY UVESH SHAIKH

## Week 2 Report: Text Vectorization and Complaint Classification Model

## 1. Introduction

In Week 2, the focus of the project was to build a machine learning model capable of automatically classifying citizen complaints into different categories. This was achieved by converting textual data into numerical form and applying classification algorithms.

## 2. Feature Selection

The input feature for the model is the cleaned text data (clean_text), which was prepared in Week 1. The target variable is the Complaint Type, which represents different categories of complaints.

```
[17]: ##week-2 task
      X = df['clean_text']
      y = df['Complaint Type']
```

## 3. Text Vectorization (TF-IDF)

Since machine learning models cannot understand text directly, the text data was converted into numerical format using TF-IDF (Term Frequency-Inverse Document Frequency). This technique assigns importance to words based on their frequency.

```
[18]: from sklearn.feature_extraction.text import TfidfVectorizer

      vectorizer = TfidfVectorizer(max_features=5000) ##Convert Text → Numbers

      X_vectorized = vectorizer.fit_transform(X)
```

## 4. Data Splitting

The dataset was divided into training and testing sets to evaluate the model’s performance. 80% of the data was used for training and 20% for testing.

```
[20]: from sklearn.model_selection import train_test_split##Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(
    X_vectorized, y, test_size=0.2, random_state=42
)
```

## 5. Model Training

A Logistic Regression model was used for classification. This algorithm is simple, efficient, and suitable for text classification tasks.

```
: ##Train Model (Logistic Regression)
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```

```
: LogisticRegression

LogisticRegression()
```

## 6. Model Evaluation

The model's performance was evaluated using accuracy score and classification metrics such as precision, recall, and F1-score.

```
[23]: y_pred = model.predict(X_test)
```

```
[24]: from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

Accuracy: 0.9999545681704602
```

## 7. Confusion Matrix Visualization

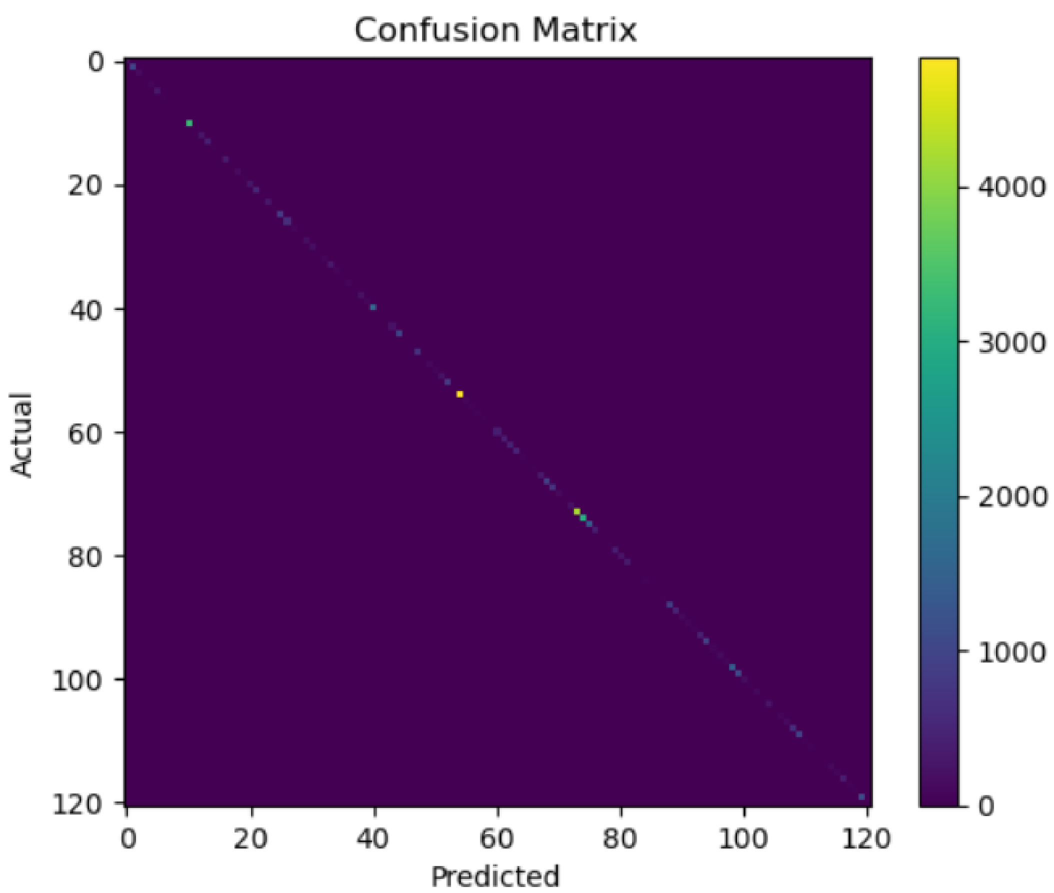
A confusion matrix was generated to visualize the model's performance by comparing actual and predicted values.

[28]:

```
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

plt.imshow(cm)
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.show()
```




## 8. Key Insights

- TF-IDF effectively converted text data into numerical features
- Logistic Regression performed well for classification
- The model was able to correctly predict most complaint categories
- Evaluation metrics helped in understanding model performance

## 9. Conclusion

In Week 2, a machine learning model was successfully developed to classify citizen complaints. Text data was transformed using TF-IDF, and a Logistic Regression model was trained and evaluated. The results demonstrate that machine learning can effectively automate complaint categorization, forming a strong base for further improvements in sentiment analysis and model optimization in the next phase.